

PROJECT PROPOSAL: OFFLINE BANGLA HTR VIA CRNN-CTC

Target Domain: Document AI & Pattern Recognition | Architecture: VGG-BiLSTM-CTC | Date: September 2026

1. Executive Summary

Handwritten Text Recognition (HTR) for Bengali presents acute technical challenges due to structural complexities: the continuous horizontal upper line (*matra*), overlapping vowel diacritics (*kar*), consonant modifiers (*fo/a*), and complex conjunct characters (*juktakkhor*). Traditional OCR relying on explicit character segmentation fails systematically on cursive handwriting due to broken strokes and irregular kerning.

This project builds an **end-to-end, offline Bangla Handwritten Text Recognition system** utilizing a **Convolutional Recurrent Neural Network (CRNN)** trained with **Connectionist Temporal Classification (CTC) loss**. The model bypasses discrete character segmentation, mapping cropped handwritten word images directly to UTF-8 Bengali text sequences, designed as an ingestion tool for downstream RAG and research agents.

2. Problem Formulation & Challenges

- Absence of Discrete Boundaries:** Glyphs connect organically along the top *matra* bar and lower loops, rendering explicit bounding-box segmentation fragile.
- High Grapheme Cardinality:** Bengali consists of 11 vowels, 39 consonants, 10 diacritics, and over 250 compound ligatures, making single-character classification impractical.
- Non-Linear Script Order:** Visual glyph placement does not always mirror phonetic storage order (e.g., *E-kar* appears visually to the left of the base consonant it modifies).

3. Proposed Architecture & Theoretical Basis

Input Image (1 x 32 x W)

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[Stage 1: VGG-Style CNN Backbone] ■■■ 7 Conv Layers + BatchNorm + LeakyReLU (H reduced: 32 -> 1, W -> W/4)

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[Stage 2: 2-Layer Bidirectional LSTM] ■■■ Captures bidirectional grapheme dependencies (T = W/4 time steps)

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[Stage 3: Linear Projection Layer] ■■■ Projects hidden states to vocabulary distribution + CTC Blank

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[Stage 4: CTC Loss / Best-Path Decode] ■■■ Sums over all alignment paths; collapses duplicates & removes blanks

- Why CNN Backbone:** Extracts hierarchical, shift-invariant visual features (strokes, loops) robust to ink variations and slant.
- Why Bidirectional LSTM:** Bengali diacritics modify preceding or trailing base characters; bidirectional passes evaluate both directions before committing to predictions.
- Why CTC Loss:** Eliminates manual character-level boundary labeling by marginalizing over all valid sequence alignments during optimization.

4. Dataset & Preprocessing Strategy

- **Target Benchmark: BanglaWriting** (Mridha et al., 2021) comprising 260 writers, 21,234 word instances, and 5,470 unique vocabulary words with bounding-box ground truths.
- **Image Normalization:** Grayscale conversion, CLAHE contrast equalization, aspect-ratio preserved resizing to fixed height $H = 32$ pixels, and zero-padding up to width $W = 128 / 256$ pixels.
- **Data Augmentation:** Random affine shear ($\pm 7^\circ$), mild perspective transforms, and morphological dilation/erosion to simulate varying fountain pen nibs and writing speeds.

5. Evaluation Metrics & Target Benchmarks

Metric	Formula / Definition	Success Threshold
Character Error Rate (CER)	$CER = (S + D + I) / N$ (Levenshtein edit distance)	$\leq 12.0\%$ on test split
Word Error Rate (WER)	Exact whole-word transcription match percentage	$\leq 28.0\%$ on test split
Inference Latency	Wall-clock time per cropped word evaluation	< 45 ms on standard CPU

6. Resource Requirements & Tech Stack

- **Framework:** PyTorch 2.x (torch.nn.CTCLoss), Torchvision
- **Computer Vision:** OpenCV (cv2), Pillow (PIL), NumPy
- **Evaluation:** editdistance, jiwer
- **Compute Requirements:** 1x NVIDIA T4 / RTX 3060 GPU (~3 hours training) or free Google Colab tier.